FEDERAL SPECIFICATION

BLEACHERS, PORTABLE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers two types of prefabricated portable bleachers.

1.2 Classification.

1.2.1 *Types.* Bleachers shall be one of the following types, as specified (see 6.1 and fig. 1 for illustrations).

Type I—Modular construction.
Type II—Beam construction.

1.2.2 *Seating capacity.* Unless otherwise specified (see 6.1), bleacher units shall be of one of the following seating capacities, as specified, (see 6.1):

Size 60—having a seating capacity of not less than 60.
Size 120—having a seating capacity of not less than 120.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

QQ-S-781—Steel, Strapping, Flat.
TT-W-572—Wood-Preservative; Water Repellent.
PPP-B-601—Boxes, Wood, Cleated-Plywood.
PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-P-116—Preservation, Methods of.
MIL-T-704—Treatment and Painting of Material.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.
MIL-STD-130—Identification Marking of U. S. Military Property.

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MIL-STD-447—Definitions of Interchangeable Substitute and Replacement Items.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

U. S. Department of Commerce Publication:

16—Lumber, American Lumber Standards for Softwood Lumber.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C., 20402.)

American Society for Testing and Materials (ASTM) Publications:

A7—Steel for Bridges and Buildings.
A36—Structural Steel.
A242—High Strength Low Alloy Structural Steel.
A307—Low-Carbon Steel Externally and Internally Threaded Standard Fasteners.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa., 19103.)

International Conference of Building Officials (ICBO): Code:

Uniform Building Code.

(Application for copies should be addressed to the International Conference of Building Officials, 610 South Broadway, Los Angeles 14, California.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified (see 6.1), a preproduction sample shall be furnished. The preproduction sample is defined as a production sample which will be identical to the product which the manufacturer produces in fulfillment of the contract. Prior to submission of any complete bleacher for final Government acceptance, the contractor shall furnish a preproduction sample for the purpose of determining conformance to the provisions of the contract and this specification. The preproduction sample shall be comprised of or fabricated from the same type or kind of material, produced by the same methods, and shall be fully representative of the planned production units in all details and respects. Approval of the preproduction sample by the Government shall not relieve the contractor of the responsibility to supply bleachers conforming to the contract or this specification. No changes or deviations from the preproduction sample will be acceptable without written approval of the contracting officer.

3.2 Material. Material shall be as specified herein. Material not definitely specified shall be of good commercial quality entirely suitable for the purpose intended. Material shall be free from all defects and imperfections that might affect the serviceability of the finished product.

3.2.1 *Structural steel.* Steel for structural members shall conform to ASTM A7, A36, or A242.

3.2.2 *Tubular steel.* Steel tubing for the tubular members shall be structural grade, high carbon, steel tubing; yield strength not less than 50,000 pounds per square inch (p.s.i.); tensile strength not less than 75,000 p.s.i.; elongation in 2 inches (in.), 15 percent. Tubing of alloy steel may be used provided that the strength and corrosion resistance are equivalent or superior.

3.2.3 *Lumber.* Lumber shall be dense, select, structural grade, West Coast, Douglas

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fir or dense, structural grade, southern, yellow pine. All lumber shall be kiln dried to a moisture content not greater than 16 percent. Lumber shall be surfaced four sides to size standards of the Department of Commerce Simplified Practice Recommendation 16.

3.3 Construction. Bleachers shall be furnished unassembled, complete with wood seatboards, footboards, and steel supporting framework, as specified herein. Seating capacity of bleacher unit shall be based on a width per seat of not less than 18 in. nor more than 19 in. Modular construction, type I, shall have a supporting framework of 3, 4, or 5 seat row modules, and the multiple row modules are joined to form the full depth of the bleacher for the required capacity. Beam construction, type II, shall have supporting beams or stringers extending full depth of the bleacher unit at right angles to the seat rows. Supporting modular frames or stringers shall be spaced not more than 6 feet center to center. Supporting members shall be cross-braced throughout the entire structure in a manner to adequately meet the design load requirements. All fasteners, straps, yokes, and plates shall be furnished complete by the supplier. Bleachers shall be regular commercial items which meet the following minimum requirements:

(*a*) Height to top of first seat,	not to exceed 18 in.
(*b*) Depth between seat rows,	not less than 22 in. nor more than 24 in.
(*c*) Rise per row,	not less than 8 in. nor more than 10 in.
(*d*) Vertical distance from footboard to corresponding seatboard,	not less than 16 in. nor more than 18 in.
(*e*) Live load per lineal foot of seatboard and of footboard,	not less than 120 pounds (lb.)
(*f*) Horizontal sway load, applied lengthwise to seats, per lineal foot,	not less than 24 lb.
(*g*) Horizontal sway load, applied perpendicular to seats, per lineal foot,	not less than 10 lb.
(*h*) Guard rails at ends and back, horizontal load at top rail, per lineal foot,	not less than 20 lb.
vertical load on top rail, per lineal foot,	not less than 75 lb.

Unless otherwise specified (see 6.1), steel guard rails shall be furnished on ends and back of bleacher at locations where top of seatboards is more than 4 feet above grade, (see 3.8). Unless otherwise specified (see 6.1), wood sleepers for bolting to the base of the framework or supports shall be furnished.

3.4 Fasteners.

3.4.1 *Bolts and nuts.* Bolts and nuts shall be zinc coated and shall conform to ASTM A307.

3.4.2 *Flat washers.* Flat washers shall be commercial grade, zinc coated.

3.5 Framework. Structural steel members shall be shop drilled, or subpunched and reamed for field connections and fasteners. Steel tubing members shall be electrically welded.

3.6 Seatboards and footboards. Seatboards and footboards shall be cut to design length from not less than 2 in. by 10 in. nominal planking, and shall be jig drilled and relieved for bolting of other connection to the supporting structure. Seatboards shall butt flush, and have edges eased. Exposed wood surfaces shall be free from slivers and pitch.

3.7 Ground sills. Ground sills or sleepers shall be not less than 2 in. by 6 in. nominal plank for bolting to the bleacher framework. Sills shall be treated with pentachlorophenol preservative in accordance with TT-W-572 after cutting to finished size.

3.8 Guardrail. Steel guardrail shall be structural, welded or bolted angle, or steel tubing with welded connections. The guard-

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rail shall consist of two rails with the top rail at 42 in. above the footboards for side rails, and 42 in. above seatboards for rear rail in order to comply with ICBO Uniform Building Code. Clear opening in railings shall not be greater than 18 in. All surfaces and connections shall be ground smooth, corners chamfered, and free of possibility of snagging.

3.9 Cleaning, treatment, and painting. All exposed surfaces, except zinc-coated and threaded surfaces, shall be cleaned, treated, and painted in conformance to MIL-T-704, type A. Finish color shall be as specified (see 6.1).

3.10 Interchangeability. Components having the same part number or identification mark shall be interchangeable according to the definition of MIL-STD-447.

3.11 Identification marking. The bleacher shall be marked for identification in accordance with MIL-STD-130.

3.12 Erection instructions. One set of erection instructions, including drawings or diagrams and bill of material, shall be furnished by the supplier with each bleacher; clearly indicating the location of each piece by appropriate designation.

3.13 Workmanship.

3.13.1 *Steel fabrication.* Steel used in the fabrication of components shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the metal. Shearing and chipping shall be done neatly and accurately. Reentrant cuts shall be made in the best possible manner. Flame cutting, using a tip suitable for the thickness of the metal, may be employed instead of shearing and sawing. Burned surfaces of flame-cut material shall be ground or machined sufficiently to remove slag and cracks. All bends of a major character shall be made with controlled means in order to insure uniformity of size and shape. Precautions shall be taken to avoid overheating, and heated metal shall be allowed to cool slowly. Bolt holes shall be accurately punched or drilled and shall have the burrs removed.

3.13.2 *Welding.* The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. All weld spatter and projections on mating surfaces where members are in contact for fabrication or assembly shall be ground to a suitable flat surface. Spot, tack, or intermittent welds for strength will not be permitted. Weld penetration shall be such as to provide transference of maximum design stress through the base metal juncture. Fillet welds shall be provided when necessary to reduce stress concentration. Manual and machine welding processes and materials shall be in accordance with a nationally recognized welding code.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling for lot acceptance.

4.2.1 *Inspection lot.* All bleachers offered for delivery at one time shall be considered a lot for purposes of inspection.

4.2.2 *Sampling for examination.* A random sample of bleachers shall be selected from each lot offered to the Government in accordance with MIL-STD-105 at inspec-

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tion level S-2. The acceptable quality level shall be 6.5 defects per hundred units.

4.3 **Examination.** The preproduction sample and each sample selected in accordance with 4.2.2 shall be examined for compliance with the requirements specified in section 3. Any redesign or modification of the supplier's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual and tactile examinations and dimensional measurements. If the number of defects in any sample exceeds the acceptance number for that sample the lot represented by the sample shall be rejected.

4.4 **Tests.** A preproduction sample of a completed bleacher unit shall be selected as a test erection unit, which shall be erected on a level base. The parts shall fit together without modification, forcing, or racking. Interchangeability shall be determined by transposing corresponding members. The contractor shall provide all labor, facilities, and supplies necessary for assembly and disassembly, and shall repair, including touchup painting, or replace any parts damaged as a result of this test. The test erection unit may be used as a completed production unit.

4.5 **Preparation for delivery inspection.** The preservation, packaging, packing, and marking of the bleachers shall be inspected to verify conformance to the requirements of section 5.

5. PREPARATION FOR DELIVERY

(For civil agency procurement, the levels of preservation, packaging, and packing used are defined in Fed. Std. No. 102.)

5.1 **Preservation and packaging.** The preservation and packaging shall be level A or C, as specified (see 6.1).

5.1.1 *Level A.* The bolts, nuts, and washers shall be preserved in accordance with MIL-P-116, method IC-2 or IC-5.

5.1.2 *Level C.* The complete bleachers shall be preserved and packaged in accordance with the supplier's standard practice.

5.2 **Packing.** The packing shall be level A, B, or C, as specified (see 6.1).

5.2.1 *Level A.* The bolts, nuts, washers and any other small components not suitable for bundling shall be packed in close-fitting boxes conforming to PPP-B-601, overseas type, or PPP-B-621, class 2. Where practicable, the boxed items shall be secured to the bundles. The structural members, guardrails, ground sills, seatboards, and footboards shall be bundled into secured lifts with flat steel strapping and bolting in combination with suitable wood blocking or battens, as required to form nonshifting bundles. The strapping shall conform to QQ-S-781, type I, class B, and be spaced not to exceed 36 inches on center, with end strapping placed not more than 18 inches from each end. The strapping shall be stapled to any wood blocking or battens provided.

5.2.2 *Level B.* The bleachers shall be packed as specified for level A, except that the boxes shall be domestic type or class 1, as applicable and strapping shall be either class A or B.

5.2.3 *Level C.* The complete bleachers shall be packed for shipment to insure arrival at destination in satisfactory condition and carrier acceptance at lowest rates in compliance with rules and regulations of carriers as applicable to the mode of transportation.

5.3 **Marking.**

5.3.1 *Civil agencies.* Shipments shall be marked in accordance with Fed. Std. No. 123.

5.3.2 *Military agencies.* Interior packages, shipping containers, and bundles shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specfication.
(*b*) Type of bleacher required (see 1.2 and 1.2.1).
(*c*) Seating capacity required (see 1.2.2).
(*d*) When preproduction sample is not required (see 3.1).
(*e*) When steel guard rails are not required (see 3.3).
(*f*) When wood sleepers, or sills, are not required (see 3.3).
(*g*) Color of finish paint (see 3.9).
(*h*) Level of preservation, packaging, and packing required (see 5.1 and 5.2).

6.2 Information. When aisles, stairs, or elevated construction are desired, such features should be specified in accordance with appropriate guides such as the Uniform Building Code.

6.3 Transportation description. The transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Bleachers, steel or wood, knocked down, not otherwise indexed by name.
Carload minimum weight 24,000 pounds, subject to Rule 34, Uniform Freight Classification.

Motor:

Bleachers, steel or wood, knocked down, not otherwise indexed.
Truckload minimum weight 36,000 pounds, subject to Rule 115, National Motor Freight Classification.

CUSTODIANS:

Army—MO (ERDL)
Navy—YD
Air Force—69

Review activities:

Navy—YD
Air Force—69
DSA—CS

User activity:

Army—MO

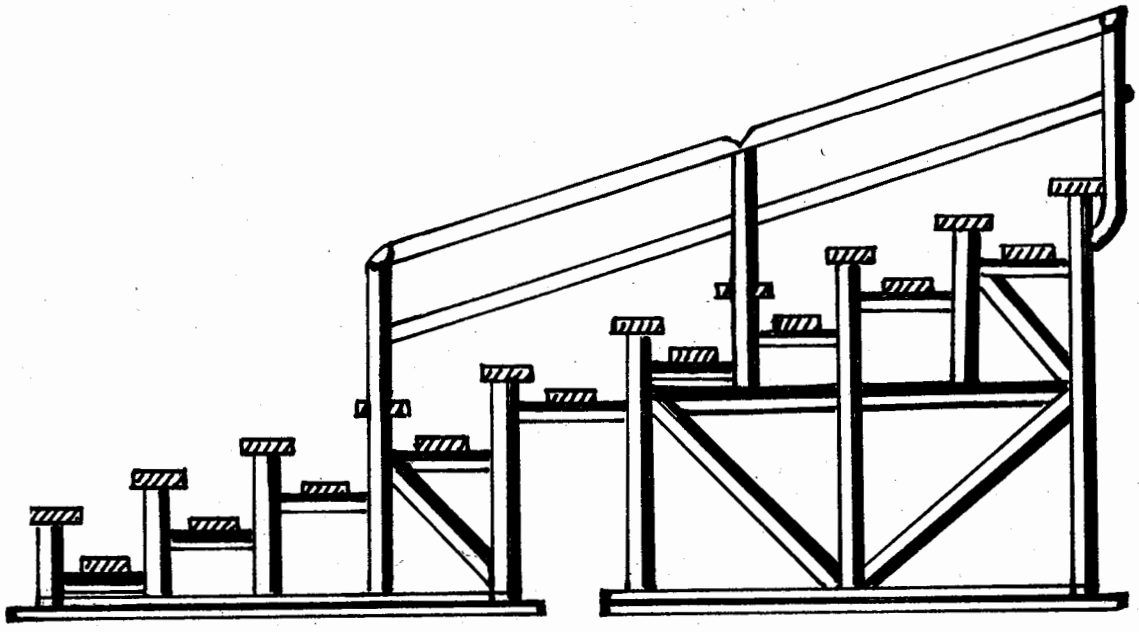
Preparing activity:

Navy—YD

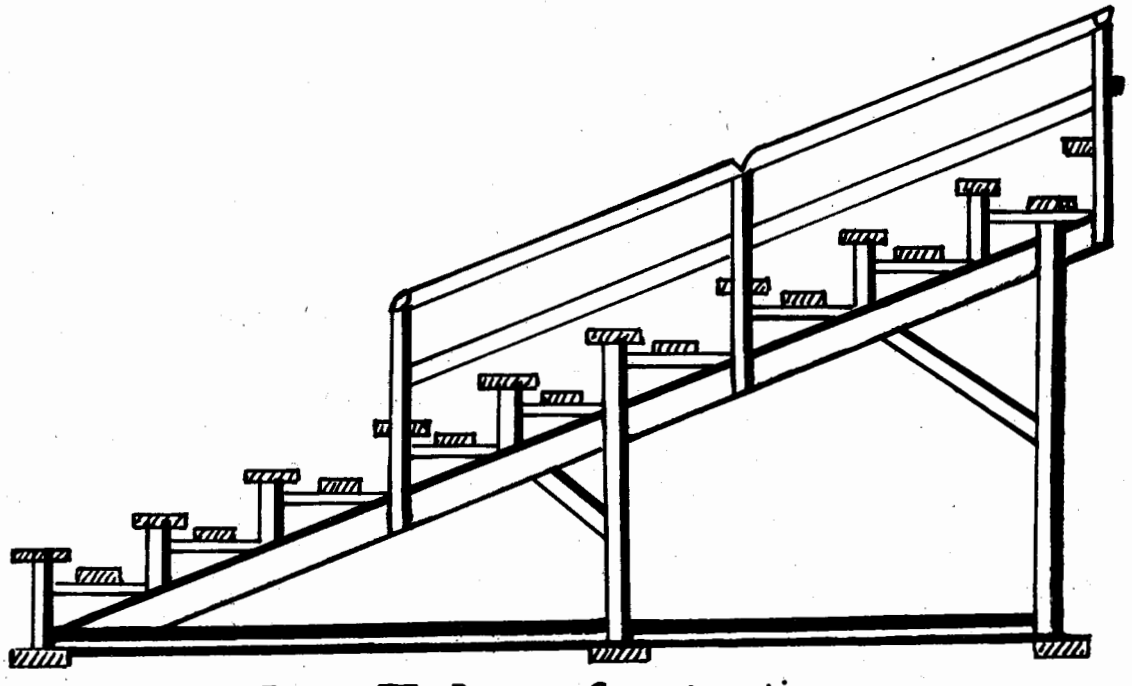
CIVIL AGENCY INTEREST:

Department of Agriculture
District of Columbia
Department of Health, Education, and Welfare
Department of the Interior
Department of Justice

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Type I, Modular Construction



Type II, Beam Construction

FIGURE 1. Portable bleachers—end view

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